

Instructor: Date:

Learner: Regional Branch:

Program: Center:

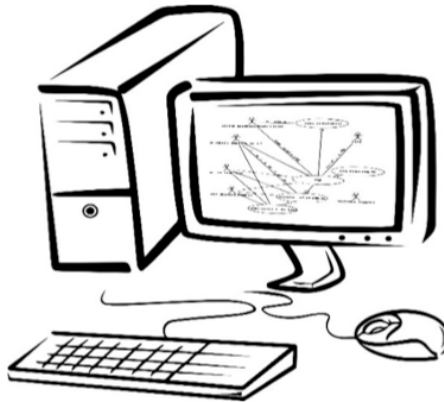
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Let's Start!

Design and Diagramming



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can say what object diagrams help visualize.
- I can talk about the elements that appear in an object diagram.
- I can discuss how to suppress elements in an object diagram.

Ask your partner:

- What are diagrams useful for?
- What information do you usually include in a diagram?





Let's Explore!

Sarah needs to organize the data she collected from her customers.



Instructions: Read the text below.

Sarah: Hey, Josh. I need your help. As you know, I collected important data about my customers; So, I need you to create a mechanism that helps me organize that data.

Josh: Sure, Sarah. I just need to create an object diagram. It will help you to have not only general but also detailed information about them.

Sarah: What exactly can I see on an object diagram?

Josh: Well, object diagrams help us visualize one instance of time and all the data that the system has in that specific moment. They are common to illustrate particular snapshot scenarios.

Sarah: That sounds great. But, I usually don't want to overload my system with a bunch of data that I don't need. Also, my customers sometimes don't want to provide certain information.

Josh: That's not an issue. Object diagrams are very useful; they look similar to class diagrams but they don't need to have an instance specification.

Sarah: What kind of elements do we need to include in an object diagram?

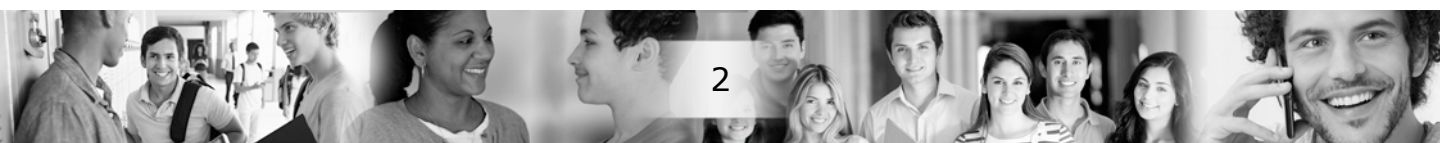
Josh: First, you need an object name and class name to describe the elements on the diagram. For instance, the object name of your diagram could be "MyFrequentCustomers" and the class name could be "Customers".

Sarah: Is that it?

Josh: Not really, we also need to include attributes and their values. The names of the customers could be the attributes and their contact information, the values.

Sarah: And, what if I don't have the customer's name but I have his contact information? , or what if I don't want to include certain data into the object diagram? Can I do that?

Josh: Definitely! That is what we call suppressing elements. We can suppress some attributes, or all of them if we want to. We can even suppress the object name. The diagram will only show us what we want to see.





Let's Practice!

Sarah sometimes doesn't want to include all her clients' information.

Instructions: Circle the correct word/s to complete the paragraph.

My customers sometimes (don't / doesn't / did) want to provide all their contact information. So, I need a diagram where I can suppress elements. I (usual / use / usually) don't (wants / want / wanted) to see all the data about a single customer because most of the times I look for specific information; for that reason, I always use object diagrams.



Let's Practice!

What does Sarah need to do?

Instructions: Proofread the sentences. Correct the mistakes.

1. Sarah needs to use an object diagram. Her collected a lot of data.

2. Sarah puts her customers' data on an object diagram. She really cares about they.

3. Object diagrams help visualize one instance of time and all the data that the system has in that specific moment. Them are common to illustrate particular snapshot scenarios.



Extension Activity

Now it's time for you to organize your data!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about UML object diagrams.
- Pick a specific situation in which you need to organize data in a diagram (like in Sarah's case). Think about your needs: Do you want to include all information? What kind of information do you need?
- Organize your data in a diagram according to your needs.
- Share it with your partners.

Self-Assessment

Yes No

- I look for more information about UML object diagrams. ☐ ☐
- I think about particular needs when organizing data in diagrams. ☐ ☐
- I organize data in a diagram according to those needs. ☐ ☐
- I share my results with my partners. ☐ ☐



Target Words

- **instance** (noun): example of a situation
e.g. Object diagrams help us visualize one instance of time and all the data that the system has in that specific moment.
- **snapshot** (noun): image of the current status of something
e.g. They are common to illustrate particular snapshot scenarios.
- **instance specification** (noun): instance you describe with proper detail
e.g. They look similar to class diagrams but they don't need to have an instance specification.
- **visualize** (verb) : to see; observe
e.g. Object diagrams help us visualize one instance of time and all the data that the system has in that specific moment.
- **overload** (verb) : to put an extra load on something and exceed its capacity
e.g. I usually don't want to overload my system with a bunch of data that I don't need.



Learning Points

Want to in Negative Statements with Adverbs of Frequency

You use the verb **want to** plus another verb to express wishes.

You can use **want to in negative statements** to express that you do not wish that something happens. In order to create these type of sentences, follow the next structure:

Subject + **do not (don't) / does not (doesn't) + want to** + verb + complement

Example:

We **don't want to** include too much information on the diagram.

Adverbs of Frequency

We can complement the previous sentences by expressing the frequency in which you **don't want** something to happen. These are **adverbs of frequency** that you can use for these statements:

Examples

- *I usually don't want to overload my system with a bunch of data that I don't need.*
- *My customers sometimes don't want to provide certain information.*



Learning Points

Subject and Object Pronouns

You use **subject** and **object pronouns** to replace subject or objects within a statement.

We use **subject pronouns** when we want to reduce proper and common nouns repetition within statements.

The following charts shows you what the **subject** and **object pronouns** are:

Subject Pronouns	Object Pronouns
I	Me
You	You
He	Him
She	Her
It	It
We	Us
They	Them

Let's see how we can use **subject pronouns** to avoid repetition by comparing these two group of sentences:

The engineer designs a new software. **The engineer** needs to visualize the data so, **the engineer** uses a diagram.

The engineer designs a new software. **He** needs to visualize the data so, **he** uses a diagram.

Now, let's see how we can use **object pronouns** to avoid repetition by comparing these two group of sentences:

The engineer designs a new software for **his boss**. He really wants to impress **his boss**.

The engineer designs a new software for **his boss**. He really wants to impress **him**.

Examples

- I just need to create an object diagram. It will help you to have not only general but also detailed information about them.
- Object diagrams help us visualize one instance of time and all the data that the system has in that specific moment. They are common to illustrate particular snapshot scenarios.

